11

```
C:\Users\Student\Desktop\1IC23CY012>java DiffieHellman
Enter a prime number (p): 23
Enter a primitive root (g): 5
Enter Alice's private key (a): 6
Enter Bob's private key (b): 15

Alice's Public Key (A): 8
Bob's Public Key (B): 19

Shared Secret Key computed by Alice: 2
Shared Secret Key computed by Bob: 2

Key Exchange Successful! Shared key is same.
```

10

```
C:\Users\Student\Desktop\IIC23CY012>java RSA
Original Message: HELLO
Encrypted Message: 678497148983076895925733812D451509157668095630052072294161691822212117774 3
606088668146185661272440814008112455509239231632462275349878350987761485724111444918867012038 9
580193089573892489323288702889502023122340088883172095580193095788828783330377483322889885295
643410014499111786412224407266422084113899675337748111813936218749193888086657677152520632612 7
660176856072107624877248297832993390588837248337223248897733388838277348993992435557389238 88
024834155400999961791563190394307964856194608972550869725501799700953334775979955230117999032 1
033399503277323937995339820983726730933270589483283305337903317258419483202519976717517845686 12
730154314
Decrypted Message: HELLO
```

12

```
C:\Users\Student\Desktop\IIC23CY012>javac FermatEuler.java

C:\Users\Student\Desktop\IIC23CY012>java FermatEuler
Enter value of a: 3
Enter prime number p (for Fermat): 5
Enter number n (for Euler): 10

Fermat Result (a^(p-1) mod p): 1

Euler Totient ϕ(n): 4
Euler Result (a^ϕ(n) mod n): 1
```

8

```
C:\Users\Student\Desktop\IIC23CY012>java LCG
Enter seed (X0): 5
Enter multiplier (a): 3
Enter increment (c): 7
Enter modulus (m): 11
Enter number of random numbers to generate: 5

Generated Pseudo Random Numbers:
X1 = 0
X2 = 7
X3 = 6
X4 = 3
X5 = 5
```

9

```
C:\Users\Student\Desktop\IIC23CY012>java BlumBlumShub
Generated Random Bits:
1 1 1 0 1 1 1 1 1 0 0 0 1 0 1 0 1 0 0 1

Generated Random Number (16 bits):
22086
```

7

```
C:\Users\Impact\Desktop\1IC23CY012>java SDES
Enter 10-bit key: 1111000011
Enter 8-bit plaintext: 11110000
Encrypted: 01110101
Decrypted: 11110000
```